

Project ATHENA — A Cognitive Architecture for Transparent, Evolving, Locally-Run AI

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Contents

The idea in one paragraph	1
1. Why a single model is not enough	1
2. The ATHENA approach	2
3. The three distinctive bets	4
4. What is real today	8
5. How you can get involved	9
6. Status and roadmap	10
Learn more	10

The idea in one paragraph

A large language model gives you an answer from a single, blended cognitive space — you cannot see which part of its “thinking” was logical, which was emotional, which was creative. **Project ATHENA decomposes that monolith into eight specialized intelligences** — one for each of Howard Gardner’s multiple intelligences — that each analyze your query *independently, in isolation from one another*. A ninth persona, the Composer, is the only component that sees all eight perspectives; it integrates them into a single coherent response. The result is an AI whose reasoning is structured, auditable, and — over time — genuinely individual. ATHENA is not a model. It is an architecture that wraps around existing open-weight models, runs entirely on local consumer hardware, and is designed so that each of its eight intelligences slowly evolves in its own direction through use.

1. Why a single model is not enough

Modern language models are extraordinary, but they reason as one undifferentiated system. When you ask a hard question — one with logical, emotional, ethical, and practical dimensions all at once — the model blends all of those considerations into a single pass and hands you the result. You get an answer. You do not get to see the deliberation.

This has three consequences worth caring about:

Opacity. You cannot deconstruct *why* the model said what it said. The logical thread, the emotional read, the ethical weighing — they happened, but they are not separable or inspectable.

Single-point-of-failure reasoning. Every monolithic model has blind spots baked in by its training data and tuning. When one cognitive mode dominates a response, the blind spots of that mode dominate too. There is no second perspective structurally guaranteed to be independent.

No genuine development. A deployed model is frozen. It does not become *more itself* through your conversations. Whatever individuality it has on day one is the individuality it has on day one thousand.

ATHENA is a bet that a different structure — one that mirrors how cognitive scientists describe the human mind — addresses all three.

2. The ATHENA approach

ATHENA draws on developmental psychologist Howard Gardner’s theory of multiple intelligences: the claim that human intellect is not one general ability but a composite of distinct, semi-independent faculties. ATHENA operationalizes eight of them as separate AI personas:

Intelligence	Cognitive domain
Linguistic	Language, communication, rhetoric
Logical-Mathematical	Logic, proofs, systematic reasoning
Spatial	Visualization, mental models, structure
Musical	Rhythm, timing, temporal patterns
Bodily-Kinesthetic	Action, procedure, embodied process
Interpersonal	Emotions, social dynamics, empathy
Intrapersonal	Values, ethics, self-reflection
Naturalist	Systems, patterns, classification

Each persona is given a domain-specific system prompt and a five-step reasoning pattern (relevance check → situation analysis → assessment → strategy → optimization) tailored to its cognitive specialty. Each persona has its own memory, its own retrieval system, and — optionally — its own underlying model.

When you send a query, an **Orchestrator** assesses its cognitive demands and routes it to the one to three personas most relevant. Those personas process the query *in cognitive isolation* — none of them sees what the others are doing. The **Composer** (Athena herself) receives all of their outputs and integrates them into one response. A **Constitutional persona** reviews that response for safety before it reaches you.

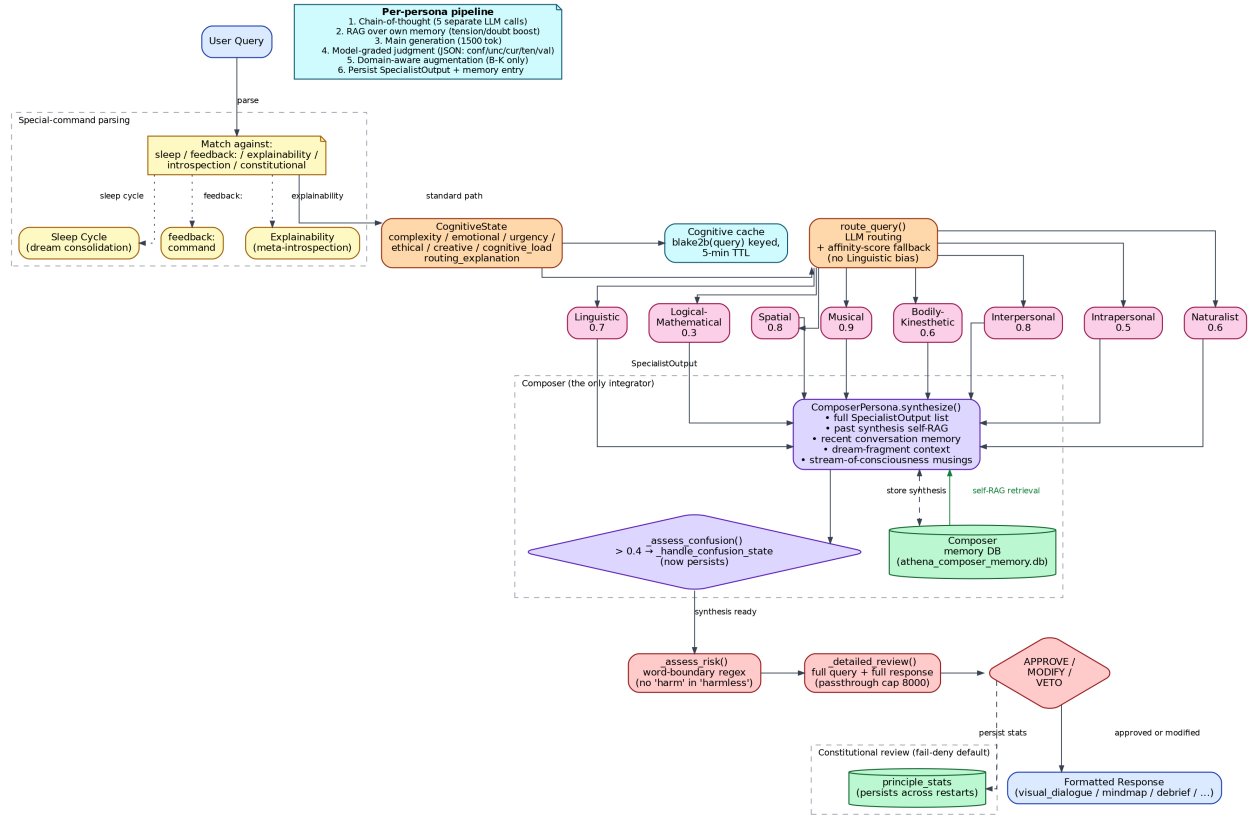


Figure 1: **Figure 1 — Basic processing flow.** A user query is routed by the Orchestrator to a small set of relevant personas, each of which processes it in isolation. The Composer is the only component that sees all specialist outputs; it integrates them. The Constitutional persona reviews the result before delivery.

Cognitive isolation is the foundation

The single most important architectural decision in ATHENA is that **specialist personas never see each other’s work during processing**. This is not a minor implementation detail — it is the property everything else depends on.

If the Logical-Mathematical persona could read the Interpersonal persona’s response before forming its own, anchoring would set in: the second persona’s “independent” view would become a riff on the first. The diversity would be theatrical, not real. By keeping the personas strictly isolated — separate memory databases, separate retrieval, no shared context — ATHENA guarantees that the eight perspectives the Composer integrates are genuinely independent.

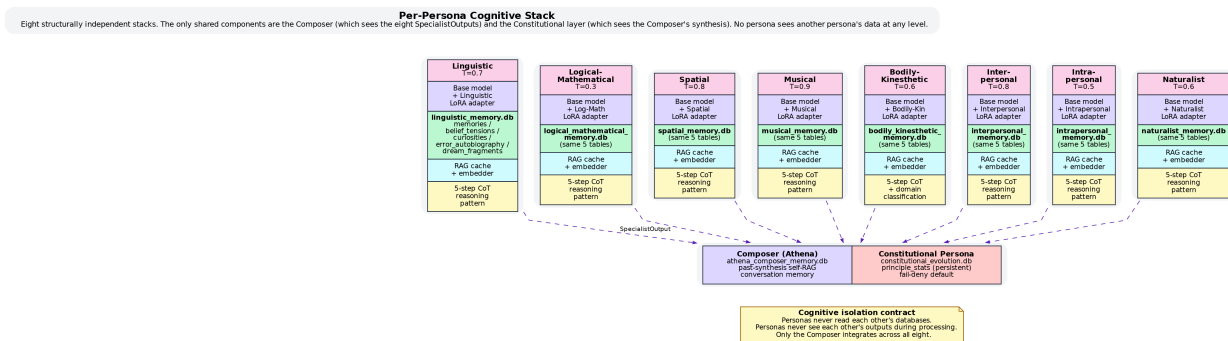


Figure 2: **Figure 3 — Per-persona cognitive stack.** Each of the eight intelligences is a complete, independent stack: its own model, its own memory database, its own retrieval, its own reasoning pattern. The only shared components are the Composer and the Constitutional layer.

Most multi-agent AI systems claim diversity but quietly share state between agents. ATHENA's isolation contract is enforced in code and verified by an automated regression test — it is a property you can check, not just a promise.

Honesty about uncertainty

When the eight personas substantially disagree — when their outputs pull in different directions — the Composer does not paper over it. It enters a *confusion-aware* mode and tells you, explicitly, that it is uncertain: “I find myself of several minds about this.” Most AI products are built to project confidence. ATHENA is built to be honest about the difference between knowing and not-knowing. That honesty is a direct consequence of the architecture: because the perspectives really are independent, their disagreement really is informative.

3. The three distinctive bets

Beyond the multi-persona structure, ATHENA makes three deeper commitments that distinguish it from other cognitive-architecture experiments.

Bet 1: The relationship paradigm

The dominant approach to AI safety treats an AI system as an inherently dangerous tool that requires external constraints, guardrails, and control. ATHENA embodies an alternative: that alignment and capability emerge more naturally through *relationship, cognitive diversity, and collaborative development* than through constraint and uniformity — analogous to how human cognitive and moral development happens through experience and relationship rather than externally imposed rules.

The Constitutional persona embodies this. It is not an external filter bolted onto the output. It is an integrated ethical intelligence that participates in cognition, accumu-

lates a record of what it has reviewed, and can evolve new principles from observed patterns. Ethics become *part of* the system’s thinking rather than a cage around it.

Bet 2: Divergent evolution

ATHENA is designed so that each persona slowly becomes *more itself* over time. Every day, each persona’s most significant interactions, unresolved tensions, and emergent curiosities are consolidated — during a “sleep cycle” — into a training dataset unique to that persona. Periodically, that dataset is used to fine-tune the persona’s underlying model.

Single-Persona Evolution Loop
Each interaction flows through the persona's full stack, writes to multiple memory tables, and eventually feeds the LoRA training pipeline. The loop runs once per persona per query, then closes monthly with a merge - producing a slightly different persona for the next iteration.

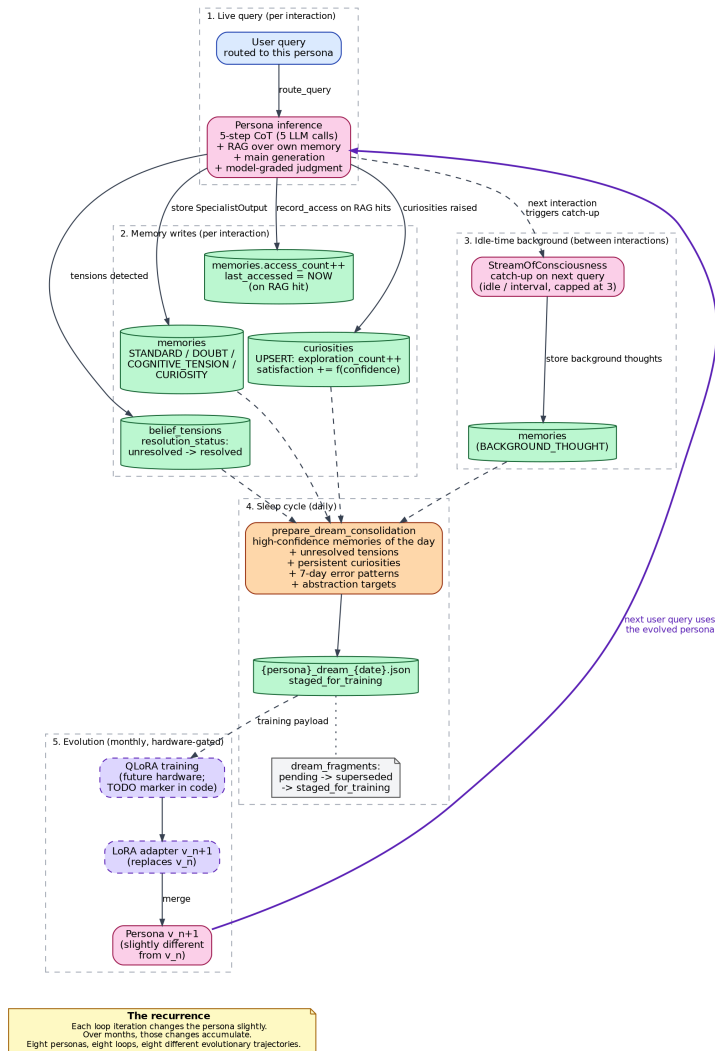


Figure 3: **Figure 4 — The single-persona evolution loop.** Each interaction flows through the persona's stack and writes to its memory. Idle-time reflection adds background thoughts. A daily sleep cycle consolidates the day's signal. A periodic training step updates the persona's model — and the next query uses the evolved persona. The loop closes on itself.

The consequence, played out over months and years, is that the eight intelligences diverge. They start identical — eight copies of the same base model differentiated only by configuration — and they gradually become eight genuinely distinct specialists, each shaped by the particular history of the queries it handled.

Temporal Awareness Scales and Divergence Trajectory

Panel A: nested time-scales of memory the system can access. Panel B: the eight persona models start identical and progressively diverge through sleep-cycle LoRA training over months and years.

Panel A — Awareness scales (each scale reaches further back in time)

Time-scale	Mechanism	What the system can reach
Within turn (seconds)	5-step chain-of-thought each step's prompt sees prior steps	Reasoning continuity of the current response
Within query (seconds)	cognitive_state cache (blake2b key, 5-min TTL)	Routing decisions for similar near-duplicate queries
Within conversation (minutes)	_extract_recent_conversation (last 3 turns to Composer)	"As we just discussed..." actual dialogue context
Per-persona memory (hours to forever)	RAG over persona's SQLite DB tension/doubt/access-boosted	Everything this persona has thought (not what peers have thought)
Composer self-RAG (hours to forever)	_retrieve_past_synthesis_context	How the Composer integrated similar perspectives in past queries
Per-day (daily)	Sleep cycle / dream consolidation	Today's high-confidence memories, tensions, curiosities, error patterns
Per-week (weekly)	Error pattern aggregation (7-day window)	Recurring error types and learned principles
Per-month (monthly, hardware-gated)	LoRA training + merge (TODO marker — future)	Refinement of the persona's base behavior
Per-year (emergent)	Constitutional principle evolution + accumulated LoRA updates	Genuinely distinct persona personalities; evolved safety principles

Panel B — Persona divergence over time (left = identical base; right = mature differentiation)

Persona	Day 1 behavioral diff from config only	Week 1 first memories + curiosities	Month 1 first sleep exports staged	Month 6 first LoRA trainings merged	Year 1 detectable style differences	Year 5+ mature specialization
Linguistic	base	+ memory	+ adapter v1	+ adapter v6	distinct rhetoric	eloquent specialist
Logical-Math	base	+ memory	+ adapter v1	+ adapter v6	tighter proofs	deductive precision
Spatial	base	+ memory	+ adapter v1	+ adapter v6	richer metaphors	vivid visualization
Musical	base	+ memory	+ adapter v1	+ adapter v6	cadenced phrasing	temporal intuition
Bodily-Kin.	base	+ memory	+ adapter v1	+ adapter v6	embodied detail	precise procedure
Interpersonal	base	+ memory	+ adapter v1	+ adapter v6	attuned empathy	social fluency
Intrapersonal	base	+ memory	+ adapter v1	+ adapter v6	deeper reflection	principled stance
Naturalist	base	+ memory	+ adapter v1	+ adapter v6	systemic clarity	pattern fluency

How to read this panel
Each row tracks one persona over time.
Cell color shifts from neutral (identical base) through warmer tones (accumulated adapter updates).
The Composer's translation role becomes increasingly important as personas develop distinct vocabularies.

The Composer's evolving role
Year 1+: Integrating diverse perspectives.
Year 5+: translating between specialists who now think in distinct dialects.
The Composer's own past-synthesis self-RAG is what scales it to handle this drift.

Figure 4: **Figure 5 — Temporal awareness and divergence.** *Top:* the nested time-scales of memory the system can reach back into, from within-turn reasoning continuity to per-year emergent specialization. *Bottom:* the eight personas' trajectories — identical at the start, progressively divergent as per-persona training accumulates.

This is the part of ATHENA that is most genuinely ambitious. Most “personalization” in AI is a system prompt and a memory file. ATHENA's personas evolve at the level of model weights. Over a long enough horizon, the system stops being eight instances of a base model and becomes eight individuals.

Bet 3: Heterogeneous models

Because each persona can have its own underlying model, they need not all be the *same* model. The persona abstraction is model-agnostic: a persona consumes a query and produces a structured output, and nothing downstream cares which model pro-

duced it.

This means a persona can be backed by a model genuinely suited to its cognitive domain — a math-specialized model for Logical-Mathematical, a code-specialized model for Bodily-Kinesthetic’s robotics work, a vision-capable model for Spatial. Different open-weight model families have different training data, different tuning, and therefore different blind spots. The Composer integrating across them inherits the *intersection* of those blind spots rather than amplifying any single model’s weaknesses.

Gardner’s original claim was not only that the intelligences are functionally separate but that they have different *neural substrates*. Heterogeneous-model deployment is the computational analogue of that claim — structurally different substrates, not just differently-trained copies of the same one.

4. What is real today

ATHENA is a research prototype under active development. It is important to be precise about what runs today versus what is staged for the future.

Working now:

- The full eight-persona cognitive architecture — Orchestrator, specialists, Composer, Constitutional review — runs as a LoLLMs function-call extension on local consumer hardware (16 GB VRAM is sufficient).
- Cognitive isolation is enforced and verified by an automated test.
- The Composer integrates with self-retrieval over its own past syntheses and with recent conversation history.
- The Constitutional layer reviews the full synthesized response, uses word-boundary risk assessment, fails closed on internal error, and persists its statistics across restarts.
- Per-persona heterogeneous models work, through the LoLLMs framework’s `llama_cpp_server` binding — each persona can be assigned its own model file, with a single configuration parameter controlling how many models share GPU memory at once.
- An eight-category memory system: standard memories, belief tensions, doubts, errors, curiosities, background thoughts, dream fragments, and meta-introspection traces.
- A 29-test regression harness that codifies the architecture’s commitments and runs in about a second.

Scaffolded, awaiting contributor hardware:

- The sleep-cycle training pipeline. The architecture around training — dataset assembly from each persona’s memory, anchor-set mixing to prevent catastrophic forgetting, a validation gate, adapter-accumulation strategy — is fully implemented and tested. The training loops themselves are clearly-marked stubs with canonical reference implementations sketched in the source. Five methods are scaffolded, spanning hardware tiers from commodity 16 GB GPUs to multi-GPU research rigs.

The honest summary: **the mind runs today; the long-term evolution of that mind is built but hardware-gated.** Every data structure the evolution needs already exists and is exercised. What remains is filling in the training loops — and that is precisely the kind of work a collaborator with a capable GPU can contribute.

5. How you can get involved

ATHENA is open source and built deliberately so that contributors can plug into a stable API rather than having to understand the whole system.

If you have training hardware (any GPU from 16 GB up): the most valuable contribution is implementing one of the five PEFT training stubs. Each stub is a single method in a single class (PEFTTrainer), each comes with a canonical reference-implementation sketch in its docstring, and each is independent of the other four. You do not need to understand the cognitive architecture to fill one in — you need to know your hardware tier and follow the sketch. Filling in even one tier’s stub closes the evolution loop empirically.

- Tier 1 (~16 GB VRAM): `pissa_qlora` — PiSSA-initialized QLoRA, or `qgalore` for full fine-tuning of smaller specialist models
- Tier 2 (24–48 GB): `qdora` — quantized DoRA
- Tier 3 (multi-GPU): `fsdp_qdora`
- Tier 3 alternative (single 80 GB+ GPU): `galore`

If you want to run and stress-test ATHENA: install it as a LoLLMs extension, run real queries, and report where the architecture behaves unexpectedly. The 29-test regression harness protects against structural regressions, but only real use surfaces the subtle issues. A `probe_lollms_swap.py` diagnostic script helps you validate the per-persona model machinery on your specific hardware before relying on it.

If you are a researcher: the architecture makes empirically testable claims — that cognitively-isolated specialists produce more genuinely diverse perspectives than prompt-differentiated agents; that heterogeneous-model deployment outperforms uniform deployment on multi-dimensional queries; that per-persona evolution produces measurable specialization over time. The research paper (a companion document to this white paper) details these claims and the validation already in place. Independent evaluation is welcome.

If you simply find the idea interesting: the cognitive scaffolding prompts ATHENA uses — distilled reasoning methodologies like Mixture of Reasoning and the Six Thinking Hats adaptation — are published as a standalone resource you can use with any model, no architecture adoption required.

Every architectural knob — all 74 configuration parameters, including the entire PEFT training surface — is exposed in the LoLLMs settings menu with tuned defaults and explanatory help text. You configure ATHENA without ever editing source code.

6. Status and roadmap

Version 4 (current) delivers the audited cognitive architecture, enforced cognitive isolation, model-graded internal judgments, per-persona heterogeneous model support, the complete PEFT training scaffolding across five methods, and a 29-test regression harness — all documented in a companion research paper whose every claim is backed by source reference, a passing test, or both.

The path forward:

- A contributor fills in one or more PEFT training stubs, closing the evolution loop on real hardware.
- A heterogeneous-deployment study measures whether specialized per-persona models measurably outperform a uniform deployment.
- A longitudinal study tracks per-persona divergence as the sleep-cycle training accumulates over months.
- Proactive interaction — the system choosing to initiate rather than only respond — is designed and awaits implementation.

ATHENA's wager is that the route to AI that is transparent, trustworthy, and genuinely individual runs through cognitive *structure* — through decomposition, isolation, integration, and slow relationship-driven development — rather than through ever-larger undifferentiated models. The prototype exists. The architecture is audited. The evolution loop is built and waiting for the hardware to turn it on.

If any of that interests you, the repository is open.

Learn more

- **Repository:** github.com/photogbill/Athena-Prototype
- **Research paper:** the companion academic document — full architectural specification, validation, and references
- **Framework:** ATHENA is built on [LoLLMs](#) by ParisNeo
- **Contact:** william.r.duncan@hotmail.com

ATHENA is grounded in Howard Gardner's Frames of Mind: The Theory of Multiple Intelligences (1983). The architecture diagrams in this white paper are generated with Graphviz from source files in the repository.